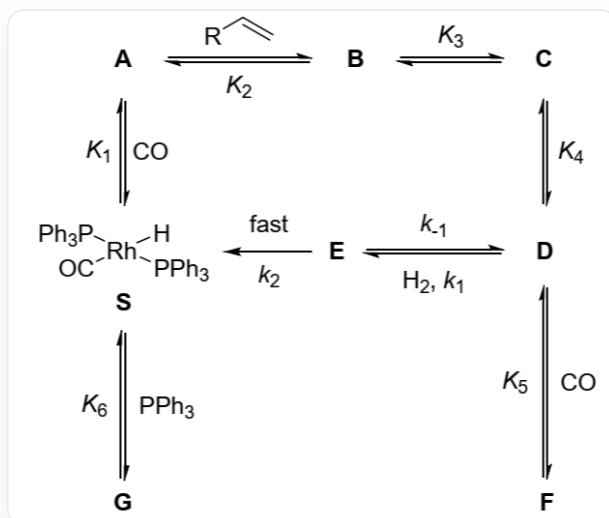


Question

The following diagram shows a kinetic model of a catalytic reaction:



The image contains nodes labeled with multiple letters: **A**, **B**, **C**, **D**, **E**, **F**, **G**, **S**. Node **S** has the molecular formula [H][Rh]([P](C1=CC=CC=C1)(C2=CC=CC=C2)C3=CC=CC=C3)([P](C4=CC=CC=C4)(C5=CC=CC=C5)C6=CC=CC=C6)[C]=O, which is the only annotation for this node. There is a two-way arrow between nodes **A** and **S**, labeled K_1 on the **S** to **A** side, and CO on the **A** to **S** side. There is a two-way arrow between nodes **A** and **B**, labeled $C=C[R]$ on the **A** to **B** side, and K_2 on the **B** to **A** side; there is a two-way arrow between nodes **B** and **C**, labeled K_3 on the **B** to **C** side; there is also a two-way arrow between nodes **C** and **D**, labeled K_4 on the **C** to **D** side. There is also a two-way arrow between nodes **D** and **E**, pointing from **D** to **E** and labeled " H_2, k_1 ", and the reverse arrow from **E** to **D** is labeled " k_{-1} ". There is a one-way arrow between nodes **E** and **S**, pointing from **E** to **S** and labeled " k_2 fast". There is a two-way arrow between **F** and **D**, labeled CO on the **D** to **F** side, and K_5 on the **F** to **D** side; and there is a two-way arrow between **G** and **S**, labeled PPh_3 on the **S** to **G** side, and K_6 on the **G** to **S** side.

The respective conversion processes in the catalytic cycle are: **S** reversibly combines with CO to generate **A**, and the equilibrium constant of the reaction is K_1 ; **A** reacts with olefin ($R-CH=CH_2$, denoted as **Aly**) to obtain **B**, and the equilibrium constant is K_2 ; **B** can be isomerized to **C**, and **C** can be isomerized to **D**, and the equilibrium constants for isomerization are K_3 and K_4 , respectively; **D** reversibly combines with CO to obtain **F**, and the equilibrium constant is K_5 . On the other hand, **D** can also react with H_2 to obtain **E**, the forward reaction rate constant is k_1 , and the reverse reaction rate constant is k_{-1} ; **E** rapidly eliminates a molecule of product **P** (not shown in the figure), and simultaneously obtains **S**, the rate constant for this step is k_2 ; **S** can also reversibly react with PPh_3 to obtain catalytically inactive **G**, and its equilibrium constant is K_6 .

It is known that the concentrations of the three species **A**, **B**, and **C** in the catalytic cycle are negligible relative to other **Rh**-containing species. The following options are descriptions of the reaction rate ($r = d[P]/dt$). Please select the correct option

by combining reasonable approximations and derivations.

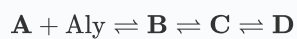
- A.** The reaction rate's order with respect to the total concentration $[\text{Rh}]$ of Rh-containing species is related to the concentrations of other species.
- B.** When the partial pressure of CO is sufficiently high, the reaction rate is -1 order with respect to $[\text{CO}]$, 1 order with respect to $[\text{H}_2]$, and depends on the values of K_1 , K_2 , K_3 , and K_4 .
- C.** When the partial pressure of H_2 is sufficiently high, the reaction rate is 0th order with respect to $[\text{CO}]$, 0th order with respect to $[\text{H}_2]$, and independent of the value of k_1 , but dependent on the value of k_2 .
- D.** When the partial pressure of H_2 is sufficiently low, the reaction rate is first order with respect to $[\text{H}_2]$, and among all the equilibrium constants and rate constants mentioned in the problem, there exists a value that is independent of the reaction rate.
- E.** When $[\text{PPh}_3]$ is sufficiently large, the reaction order with respect to $[\text{PPh}_3]$ is -1, and it is independent of $[\text{H}_2]$ and $[\text{CO}]$.
- F.** As the partial pressure of Aly continuously increases, the reaction rate will reach a maximum value that is related to $[\text{CO}]$, $[\text{Rh}]$, $[\text{H}_2]$, and $[\text{PPh}_3]$.

Answer

Correct Answer: **C**

Detailed Explanation

From the figure, we can obtain the equilibrium equations and reaction expressions including stoichiometric numbers:



CHECKPOINT

0.4 PTS

Equilibrium exists: $\text{S} + \text{CO} \rightleftharpoons \text{A}$

CHECKPOINT

0.4 PTS

Equilibrium exists: $\mathbf{A} + \mathbf{Aly} \rightleftharpoons \mathbf{B} \rightleftharpoons \mathbf{C} \rightleftharpoons \mathbf{D}$

CHECKPOINT

0.4 PTS

Equilibrium exists: $\mathbf{D} + \mathbf{H}_2 \rightleftharpoons \mathbf{E} \rightarrow \mathbf{E} + \mathbf{S}$

CHECKPOINT

0.4 PTS

Equilibrium exists: $\mathbf{D} + \mathbf{CO} \rightleftharpoons \mathbf{F}$

CHECKPOINT

0.4 PTS

Equilibrium exists: $\mathbf{S} + \mathbf{PPh}_3 \rightleftharpoons \mathbf{G}$

Based on the speculated structure and reaction process, and according to the steady-state hypothesis, give the expressions of $[\mathbf{D}]$, $[\mathbf{F}]$, and $[\mathbf{G}]$ with respect to $[\mathbf{S}]$:

$$[\mathbf{D}] = K_1 K_2 K_3 K_4 [\mathbf{S}] [\mathbf{CO}] [\mathbf{Aly}]$$

$$[\mathbf{F}] = K_1 K_2 K_3 K_4 K_5 [\mathbf{S}] [\mathbf{CO}]^2 [\mathbf{Aly}]$$

$$[\mathbf{G}] = K_6 [\mathbf{S}] [\mathbf{PPh}_3]$$

Where $[\mathbf{Aly}]$ is the concentration of the alkene species; $[\mathbf{PPh}_3]$ is the concentration of the triphenylphosphine species.

CHECKPOINT

1 PTS

The expression of $[D]$ with respect to $[S]$ is $[D] = K_1 K_2 K_3 K_4 [S][CO][Al_y]$

CHECKPOINT

1 PTS

The expression of $[F]$ with respect to $[S]$ is $[F] = K_1 K_2 K_3 K_4 K_5 [S][CO]^2 [Al_y]$

CHECKPOINT

1 PTS

The expression of $[G]$ with respect to $[S]$ is $[G] = K_6 [S][PPh_3]$

Applying steady-state approximation to $[E]$:

$$d[E]/dt = k_1 [D][H_2] - (k_{-1} + k_2)[E] = 0$$

Then:

$$[E] = k_1 [D][H_2] / (k_{-1} + k_2)$$

CHECKPOINT

1 PTS

Applying steady-state approximation to $[E]$: $[E] = k_1 [D][H_2] / (k_{-1} + k_2)$

Substituting into the material balance relationship:

$$[\text{Rh}] = [\text{S}] + [\text{D}] + [\text{E}] + [\text{F}] + [\text{G}]$$

The expression for $[\text{S}]$ is obtained as:

$$[\text{S}] = [\text{Rh}] / \{1 + K_6[\text{PPh}_3] + [1 + K_5[\text{CO}] + k_1[\text{H}_2] / (k_{-1} + k_2)] K_1 K_2 K_3 K_4 [\text{CO}] [\text{Aly}]\}$$

CHECKPOINT

1 PTS

The expression for $[\text{S}]$ is $[\text{S}] = [\text{Rh}] / \{1 + K_6[\text{PPh}_3] + [1 + K_5[\text{CO}] + k_1[\text{H}_2] / (k_{-1} + k_2)] K_1 K_2 K_3 K_4 [\text{CO}] [\text{Aly}]\}$

Substituting into the reaction rate expression:

$$r = d[\text{P}] / dt = k_2[\text{E}] = k_1 k_2 [\text{D}] [\text{H}_2] / (k_{-1} + k_2) = k_1 k_2 K_1 K_2 K_3 K_4 [\text{S}] [\text{CO}] [\text{Aly}] [\text{H}_2] / (k_{-1} + k_2)$$

Substituting $[\text{S}]$, the reaction rate expression is obtained:

$$r = k_1 k_2 K_1 K_2 K_3 K_4 [\text{Rh}] [\text{CO}] [\text{Aly}] [\text{H}_2] / (k_{-1} + k_2) \{1 + K_6[\text{PPh}_3] + [1 + K_5[\text{CO}] + k_1[\text{H}_2] / (k_{-1} + k_2)] K_1 K_2 K_3 K_4 [\text{CO}] [\text{Aly}]\}$$

CHECKPOINT

1 PTS

Reaction	rate	expression:
	$r = k_1 k_2 K_1 K_2 K_3 K_4 [\text{Rh}] [\text{CO}] [\text{Aly}] [\text{H}_2] / (k_{-1} + k_2) \{1 + K_6[\text{PPh}_3] + [1 + K_5[\text{CO}] + k_1[\text{H}_2] / (k_{-1} + k_2)] K_1 K_2 K_3 K_4 [\text{CO}] [\text{Aly}]\}$	

Judgment based on the reaction rate expression:

Regardless of the concentration values of other species, the exponent of $[\text{Rh}]$ is always 1, so the reaction order is independent of other species, therefore option A is incorrect;

CHECKPOINT

1 PTS

The exponent of $[\text{Rh}]$ is always 1, so the reaction order is independent of other species

When $[\text{CO}]$ is large enough, we have:

$$r = k_1 k_2 [\text{Rh}] [\text{H}_2] / K_5 (k_{-1} + k_2) [\text{CO}]$$

CHECKPOINT

1 PTS

When $[\text{CO}]$ is large enough, we have: $r = k_1 k_2 [\text{Rh}] [\text{H}_2] / K_5 (k_{-1} + k_2) [\text{CO}]$

The reaction rate is independent of the values of K_1 , K_2 , K_3 , and K_4 , so option B is incorrect;

When $[\text{H}_2]$ is large enough, we have:

$$r = k_2 [\text{Rh}]$$

CHECKPOINT

1 PTS

When $[\text{H}_2]$ is large enough, we have: $r = k_2 [\text{Rh}]$

Therefore, option C is correct;

When $[\text{H}_2]$ is small enough, we have:

$$r = k_1 k_2 K_1 K_2 K_3 K_4 [\text{Rh}] [\text{CO}] [\text{Aly}] [\text{H}_2] / (k_{-1} + k_2) \{1 + K_6 [\text{PPh}_3] + [1 + K_5 [\text{CO}]] K_1 K_2 K_3 K_4 [\text{CO}] [\text{Aly}]\}$$

CHECKPOINT

1 PTS

When $[\text{H}_2]$ is small enough, we have:

$$r = k_1 k_2 K_1 K_2 K_3 K_4 [\text{Rh}] [\text{CO}] [\text{Aly}] [\text{H}_2] / (k_{-1} + k_2) \{1 + K_6 [\text{PPh}_3] + [1 + K_5 [\text{CO}]] K_1 K_2 K_3 K_4 [\text{CO}] [\text{Aly}]\}$$

All rate constants and equilibrium constants in the question are related to the reaction rate, so option D is incorrect.

When $[\text{PPh}_3]$ is large enough, we have:

$$r = k_1 k_2 K_1 K_2 K_3 K_4 [\text{Rh}] [\text{CO}] [\text{Aly}] [\text{H}_2] / (k_{-1} + k_2) K_6 [\text{PPh}_3]$$

CHECKPOINT

1 PTS

When $[\text{PPh}_3]$ is large enough, we have:

$$r = k_1 k_2 K_1 K_2 K_3 K_4 [\text{Rh}] [\text{CO}] [\text{Aly}] [\text{H}_2] / (k_{-1} + k_2) K_6 [\text{PPh}_3]$$

The rate is related to $[\text{H}_2]$ and $[\text{CO}]$, so option E is incorrect.

When $[\text{Aly}]$ is large enough, we have:

$$r = k_1 k_2 [\text{Rh}] [\text{H}_2] / (k_{-1} + k_2) [1 + K_5 [\text{CO}] + k_1 [\text{H}_2] / (k_{-1} + k_2)]$$

CHECKPOINT

1 PTS

When $[\text{Aly}]$ is large enough, we have:

$$r = k_1 k_2 [\text{Rh}] [\text{H}_2] / (k_{-1} + k_2) [1 + K_5 [\text{CO}] + k_1 [\text{H}_2] / (k_{-1} + k_2)]$$

The rate is independent of $[\text{PPh}_3]$, option F is incorrect.

In summary, option C should be selected.